Numbers, Groups, Graphs, Undergrad Research, and a 14-Year-Old Conjecture

Yuval Amit

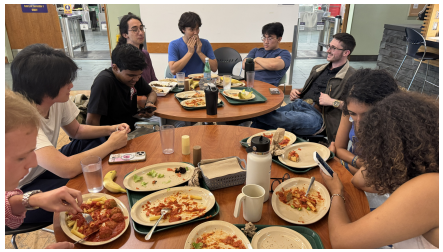
Math Club, November 2nd, 2026

Introduction

This summer, I spent 9 weeks at the SMALL Math REU.







My goal is to walk you through my process of a theorem we proved, and by the end I hope to give you a direction to continue this research!

Introduction

Let $N = \{0, 1, \ldots, n-1\}$, and $A \subseteq N$ be a subset.

$$A + A := \{x + y \mid x, y \in A\} \qquad A - A := \{x - y \mid x, y \in A\}$$

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Question

When is $|A + A| > |A - A|$?

We say that A is **MSTD** (more sums than differences) if

$$|A + A| > |A - A|,$$

MDTS if $<$, and **balanced** if $=$.

Introduction

Example (John Conway)

Let $A = \{0, 2, 3, 4, 7, 11, 12, 14\}$. Then

$$|A + A| = 26 > 25 = |A - A|,$$

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Idea:

- $x + y = y + x$, but $x - y \neq y - x$. This gives $A - A$ an advantage.
- $x + x \neq y + y$, but $x - x = y - y$. This gives $A + A$ an advantage.

The advantage for $A - A$ is bigger, so expect to see $|A - A| > |A + A|$.

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The advantage for $A - A$ is bigger, so expect to see $|A - A| > |A + A|$. A positive proportion ($> 2 \cdot 10^{-7}$) of subsets of $\{1, \ldots, n\}$ are MSTD as $n \to \infty$.

Intro to Groups (if necessary)

Definition

A **group** is a set G with a binary operation $*$ such that

1. $x * y \in G$ $\qquad \forall x, y \in G$
2. $(x * y) * z = x * (y * z)$ $\qquad \forall x, y, z \in G$
3. $\exists\, e \in G$ such that $x * e = e * x = x$ $\qquad \forall x \in G$
4. $\forall x \in G, \exists\, x^{-1} \in G$ such that $x * x^{-1} = x^{-1} * x = e$.

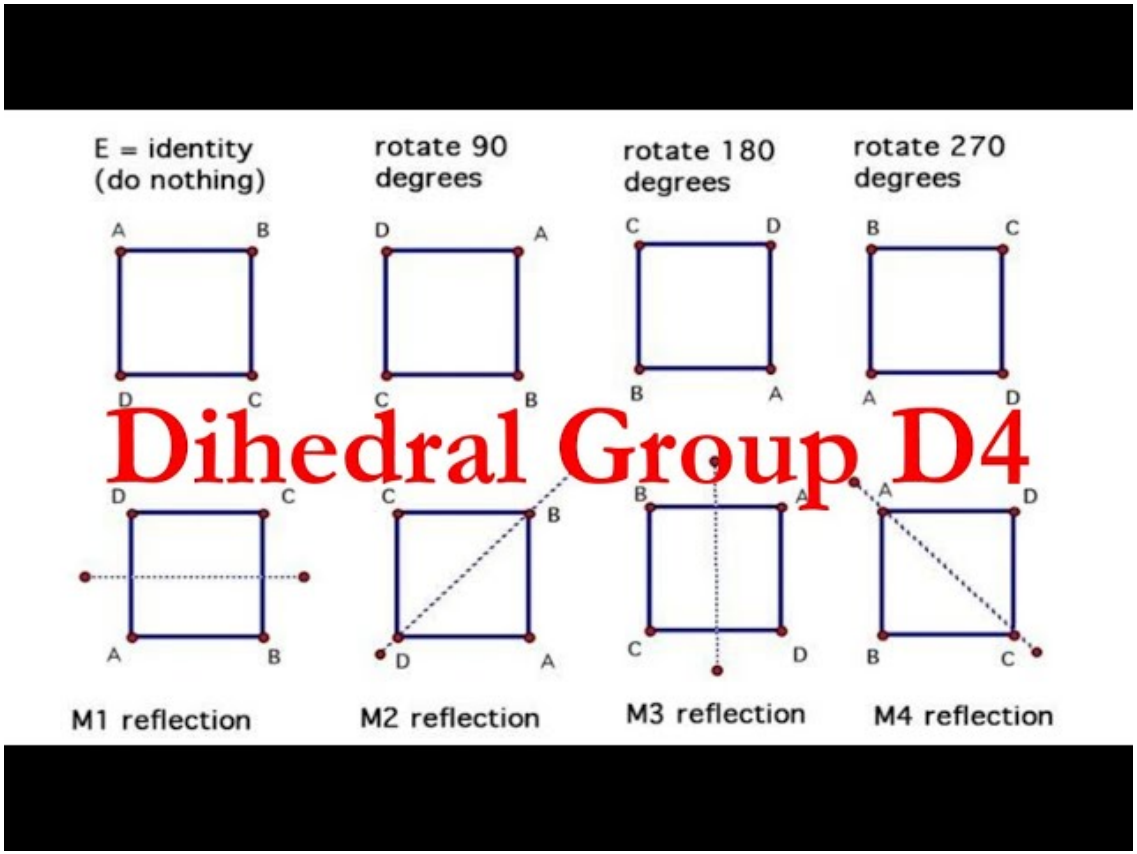
G is called **abelian** if $x * y = y * x$ for all $x, y \in G$.

Examples

$(\mathbb{Z}, +)$ is a (abelian) group, but $(\mathbb{Z}, \cdot)$ is not. The rotations of a Rubik's cube form a non-abelian group.

Intro to Dihedral Groups (if necessary)

The **dihedral group** on n vertices is the rotational/reflective symmetries of a regular n-gon.



It has $2n$ elements, and is non-abelian.

MSTD for Groups

Instead of $A \subseteq \{0, \ldots, n-1\}$, we take $A \subseteq G$ for a finite group G.

$$AA := \{xy \mid x, y \in A\} \qquad AA^{-1} = \left\{xy^{-1} \,\middle|\, x, y \in A\right\}$$

AA is called the **product set**, and AA^{-1} the **quotient set**.

Definition

We call $A \subseteq G$ **MPTQ** (more products than quotients) if

$$|AA| > |AA^{-1}|,$$

MQTP if $<$, and **balanced** if $=$.

We call G **product-dominant** if

$$|\{A \subseteq G \mid A \text{ is MPTQ}\}| > |\{A \subseteq G \mid A \text{ is MQTP}\}|,$$

quotient-dominant if $<$, and **balanced** if $=$.

Question

When is a group product-dominant or quotient-dominant?

Remark

If G is non-abelian, the quotient-set doesn't have the non-commutativity advantage. MPTQ sets will be more common.

Conjecture (Miller-Vissuet)

All dihedral groups D_n for $n \geq 3$ are product-dominant.

Computation

For a small group G, start with a score $= 0$. For each $A \subseteq G$, compute $|AA|$ and $|AA^{-1}|$.

- if $|AA| > |AA^{-1}|$ (MPTQ), add 1;
- if $|AA| < |AA^{-1}|$ (MQTP), subtract 1;
- if balanced, do nothing.

If score > 0, G is product-dominant.

If score < 0, G is quotient-dominant.

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If score > 0, G is product-dominant.

If score < 0, G is quotient-dominant.

Group	MPTQ	MQTP	Balanced	Score	Result
$\mathbb{Z}/8\mathbb{Z}$	0	64	192	-64	quotient-dominant
D_4	64	8	184	$+56$	product-dominant
Q_8	16	48	192	-32	quotient-dominant

What Makes a Group Product-Dominant?

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Is it how close G is to being abelian? No:

- $|Z(D_4)| = |Z(Q_8)| = 2$,
- they have the same number of conjugacy classes,
- *they even have the same character table!*

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Question

What property of G actually decides product- vs. quotient-dominance?

The Statistic $d(A)$

Choose $A \subseteq G$ uniformly at random, and let

$$d(A) := |AA| - |AA^{-1}|.$$

G is product-dominant if $d(A) > 0$ more often than $d(A) < 0$.

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The first two **moments** of $d(A)$ are

$$M_1(G) := \sum_{A \subseteq G} d(A), \qquad M_2(G) := \sum_{A \subseteq G} d(A)^2.$$

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Group	M_1	M_2
D_4	96	208
Q_8	-32	64

Computing M_1

Through manipulation we find

$$M_1(G) = \sum_{g \in G} \left| \left\{ A \,\middle|\, g \notin AA^{-1} \right\} \right| - \sum_{g \in G} \left| \{ A \mid g \notin AA \} \right|$$

Instead of computing $|AA|$ for all sets A, we fix g and count the sets A that *avoid* making g:

$$g \notin AA \iff \text{there are no } x, y \in A \text{ with } xy = g.$$

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Is this easier?

Forbiddance Graphs

Definition

Fix $g \in G$. The **forbiddance graphs** $\mathcal{F}_P(g)$ and $\mathcal{F}_Q(g)$ have vertex set G, with an edge $x \to y$ whenever

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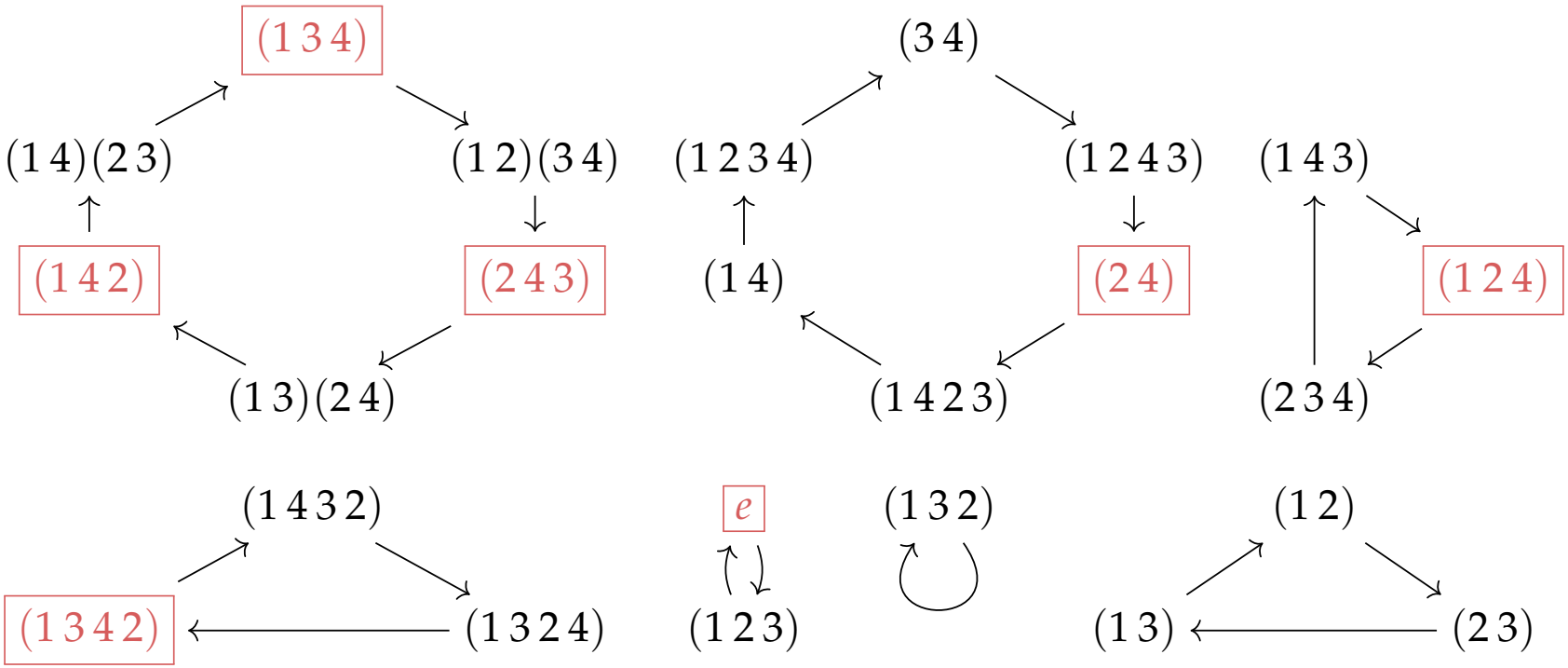
Definition

A set of vertices A is **independent** if no two vertices of A are joined by an edge. Let $i(\Gamma)$ be the number of independent sets of a graph Γ.

A is independent in $\mathcal{F}_P(g)$ exactly when $g \notin AA$, so

$$M_1(G) = \sum_{g \in G} \big(i(\mathcal{F}_Q(g)) - i(\mathcal{F}_P(g))\big).$$

Example: $\mathcal{F}_P((1\,2\,3))$ in S_4



The boxed red vertices form an independent set A, so $(1\,2\,3) \notin AA$.

Counting Independent Sets

Lemma

$\mathcal{F}_P(g)$ and $\mathcal{F}_Q(g)$ are disjoint unions of cycles.

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Remark

A cycle of length n has L_n independent sets, where L_n is the nth Lucas number ($L_1 = 1, L_2 = 3, L_{n+2} = L_{n+1} + L_n$). Also,

$$i(\Gamma_1 \sqcup \Gamma_2) = i(\Gamma_1) \cdot i(\Gamma_2).$$

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$i(\mathcal{F}_P(g))$ and $i(\mathcal{F}_Q(g))$ only depend on the cycle lengths. In the example,

$$i(\mathcal{F}_P((1\,2\,3))) = L_1 \cdot L_2 \cdot L_3^3 \cdot L_6^2 = 1 \cdot 3 \cdot 4^3 \cdot 18^2 = 62208.$$

Overview

We've translated a computation about MPTQ sets into a graph-theoretic computation:

$$M_1(G) = \sum_{g \in G} \left(i(\mathcal{F}_Q(g)) - i(\mathcal{F}_P(g)) \right)$$

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Question

How does the structure of G determine the cycle lengths?

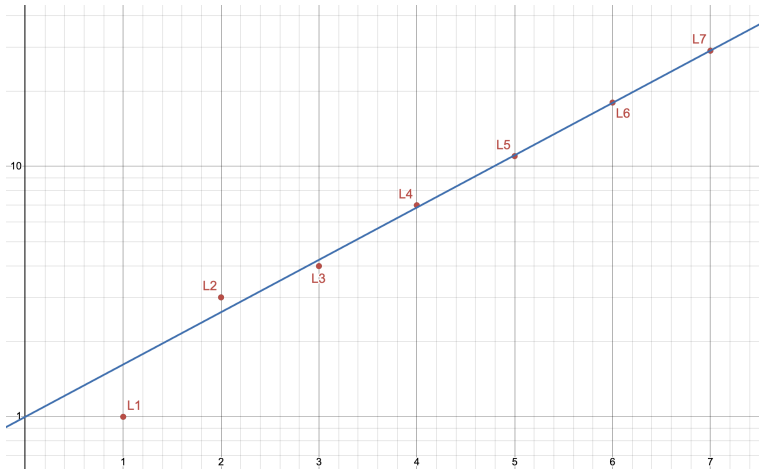
The Lucas Numbers

We write

$$i(\mathcal{F}_P(g)) = \prod_{j=1}^{m} L_{\alpha_j}, \qquad \alpha_1 + \cdots + \alpha_m = |G|.$$

If $\phi = (1+\sqrt{5})/2$ is the golden ratio, then

$$L_n = \phi^n + (1-\phi)^n \sim \phi^n.$$



Idea: L_2 is the "biggest", so will contribute the most.

2-Cycles

Now we want forbiddance graphs with lots of two cycles.

$\mathcal{F}_P(g)$: x is in a two-cycle iff

$$xy = yx = g \iff gx^{-1} = x^{-1}g = y \iff x \in Z(g)$$

$\mathcal{F}_Q(g)$: x is in a two-cycle iff

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Proposition

$\sum_{g \in G} i(\mathcal{F}_P(g))$ gets "big contributions" from central elements.
$\sum_{g \in G} i(\mathcal{F}_Q(g))$ gets "big contributions" from involutions.

The Second Moment

For M_2, we need sets A avoiding *two* elements at once.

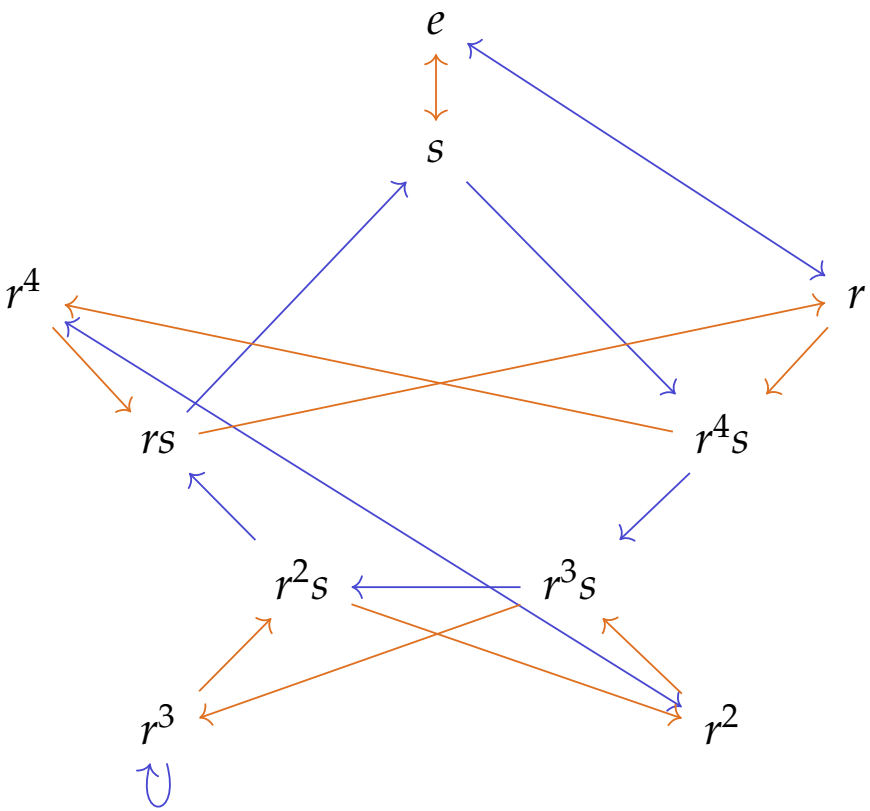
Definition

Fix $g, h \in G$. The forbiddance graphs $\mathcal{F}_{PP}(g,h)$ and $\mathcal{F}_{QQ}(g,h)$ have vertex set G, with an edge $x \to y$ whenever

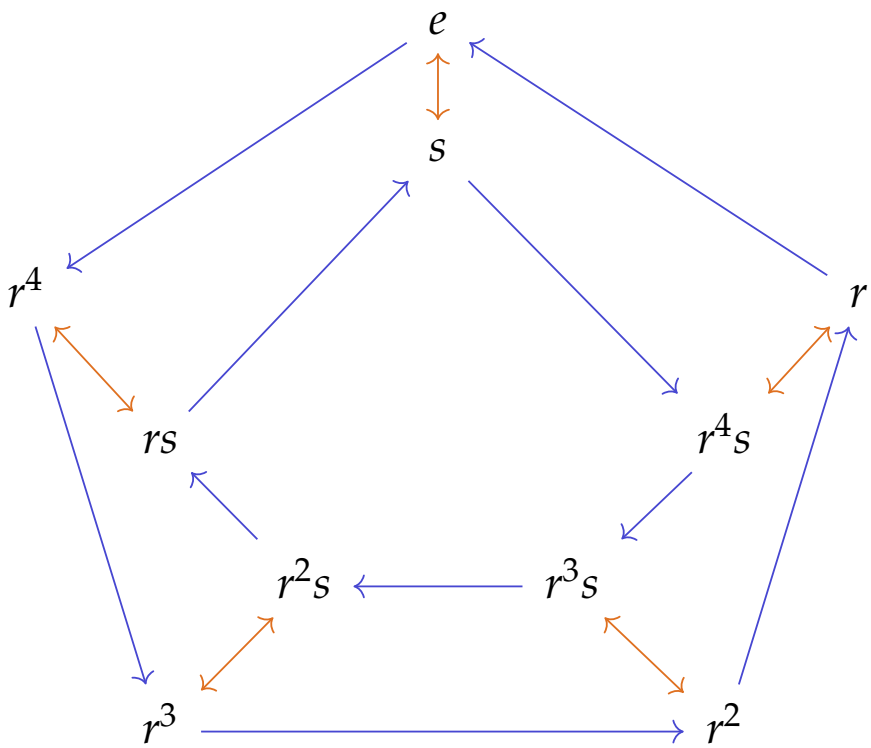
$$xy \in \{g,h\} \quad \text{in } \mathcal{F}_{PP}(g,h), \qquad xy^{-1} \in \{g,h\} \quad \text{in } \mathcal{F}_{QQ}(g,h).$$

Now A is independent in $\mathcal{F}_{PP}(g,h)$ exactly when $g, h \notin AA$.

Example: $D_5, g = r, h = s$



$\mathcal{F}_{PP}(r,s)$: 51 independent sets



$\mathcal{F}_{QQ}(r,s)$: 81 independent sets

Blue: product/quotient is r. Orange: product/quotient is s.

The Second Moment

Through manipulation we find

$$M_2(G) \leq \sum_{g \in G} i(\mathcal{F}_P(g)) + \sum_{g \in G} i(\mathcal{F}_Q(g)) + \sum_{g \neq h} i(\mathcal{F}_{PP}(g,h)) + \sum_{g \neq h} i(\mathcal{F}_{QQ}(g,h)).$$

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Remark

The double-product and double-quotient terms end up being negligible compared to the first two sums: having more edges means fewer independent sets.

The Dominance Inequality

Let B be the number of balanced subsets of G.

Theorem

Let G be a finite group, and suppose

$$\frac{M_1^2}{(2^{|G|} - B)\, M_2} > \frac{1}{2}.$$

Then

1. $M_1 > 0 \implies G$ is product-dominant,
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Example

- D_4: $\frac{96^2}{72 \cdot 208} \approx 0.62 > \frac{1}{2}$, so D_4 is product-dominant.
- Q_8: $\frac{(-32)^2}{64 \cdot 64} = 0.25$, so the theorem says nothing.

Two-Cycles Dominate

A cycle decomposition with N vertices has at most

$$\underbrace{L_2^{N/2} \approx 1.73^N}_{\text{all two-cycles}} \qquad \text{vs.} \qquad \underbrace{7^{N/4} \approx 1.63^N}_{\text{no two-cycles}}$$

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- $\mathcal{F}_Q(g)$ is all two-cycles iff g is an involution.
- $\mathcal{F}_P(g)$ is (almost) all two-cycles iff g is central.
- Everything else, including $\mathcal{F}_{PP}$ and $\mathcal{F}_{QQ}$, is exponentially smaller.

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With $T := \#\{g \mid \text{ord}(g) = 2\}$ and $Z := |Z(G)|$,

$$Q := \sum_{g \in G} i(\mathcal{F}_Q(g)) \approx T\sqrt{3}^N, \qquad P := \sum_{g \in G} i(\mathcal{F}_P(g)) \approx Z\sqrt{3}^N.$$

Back to the Dominance Inequality

Everything is now in terms of P and Q:

$$M_1 = Q - P, \qquad 2^N - B \leq Q + P, \qquad M_2 \lesssim Q + P.$$

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So

$$\frac{M_1^2}{(2^N - B)M_2} \gtrsim \frac{(Q-P)^2}{(Q+P)^2} \approx \left(\frac{T-Z}{T+Z}\right)^2.$$

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So

$$\frac{M_1^2}{(2^N - B)M_2} \gtrsim \frac{(Q-P)^2}{(Q+P)^2} \approx \left(\frac{T-Z}{T+Z}\right)^2.$$

If Z is much smaller than T (or vice versa), this is close to $1 > \frac{1}{2}$, and the sign of $M_1 \approx (T-Z)\sqrt{3}^N$ decides the winner.

- Many involutions: $g \notin AA^{-1}$ is common, so product-dominant.
- Large center: $g \notin AA$ is common, so quotient-dominant.

The Main Theorem

Theorem

Let $\{G_k\}$ be finite groups with $|G_k| \to \infty$, and let

$$T_k := \#\{g \in G_k \mid \mathrm{ord}(g) = 2\}, \qquad Z_k := |Z(G_k)|,$$

$$R_k := \#\left\{g \in G_k \,\middle|\, g^2 \in Z(G_k)\right\}.$$

1. If $Z_k = o(T_k)$, then G_k is product-dominant for large k.
2. If $T_k \sqrt{3}^{R_k/Z_k} = o(Z_k)$, then G_k is quotient-dominant for large k.

Applications

Family	Z_k	T_k	Result
D_k	1 or 2	k or $k+1$	product-dominant
S_k	1	$\geq \binom{k}{2}$	product-dominant
non-abelian simple	1	$\to \infty$	product-dominant
odd order	≥ 1	0	quotient-dominant

All for sufficiently large groups.

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All for sufficiently large groups. With a computer check of the small cases (easier said than done...), we prove Miller-Vissuet.

Future Work

- There are interesting groups with similar center size and involution counts (quaternion groups).
- The dominance inequality is not always going to work.

These avenues were left open, and I believe they're possible.

Undergrad Research

Time to yap?

Thank you!

